SIMATS ENGINEERING
ASSESSMENT TOOL 1
COURSE NAME: Artificial Intelligence COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks: 50**

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

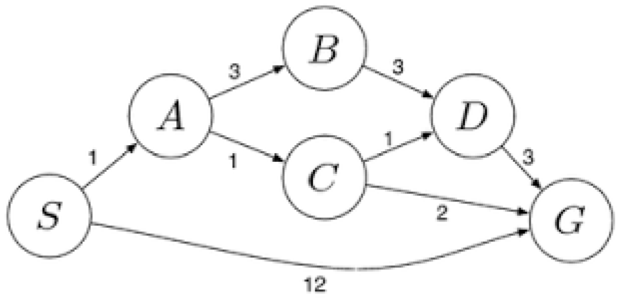
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I_Dhanush.G__(192512132)__ (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
G.Dhanush

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

1.

ANSWER: Water Jug Problem (4-Gallon Jug and 3-Gallon Jug)

- One **4-gallon jug**
- One **3-gallon jug**
- A water pump with unlimited water

Goal: Obtain **exactly 2 gallons** in the **4-gallon jug**.

State Representation

Represent each state as:

(4-gallon jug, 3-gallon jug)

Initially:

(0, 0)

Solution Steps

Step	Action	State (4-gallon, 3-gallon)
1	Fill the 3-gallon jug	(0, 3)
2	Pour 3-gallon jug into 4-gallon jug	(3, 0)
3	Fill the 3-gallon jug again	(3, 3)
4	Pour from 3-gallon jug into 4-gallon jug until the 4-gallon jug is full	(4, 2)
5	Empty the 4-gallon jug	(0, 2)
6	Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug	**(2, 0)**

After performing the above sequence of operations, the **4-gallon jug contains exactly 2 gallons of water**, satisfying the required goal state:

Goal State = (2, 0).

2.

ANSWER: **Mars Rover – Intelligent Agent Analysis**

i. What are the percepts for this agent?

Percepts are the inputs received by the Mars Rover through its sensors.

- Images from cameras
- Rock and soil composition data
- Temperature and weather readings
- Terrain and obstacle information
- Distance measurements
- Atmospheric pressure

ii. Characterize the operating environment

- **Partially observable** – It cannot observe the entire environment at once.
- **Deterministic** – Most actions have predictable outcomes, though terrain may introduce uncertainty.
- **Sequential** – Current actions affect future actions.

- **Dynamic** – Weather, dust storms, and terrain conditions may change.
- **Continuous** – Movement, time, and sensor readings are continuous.
- **Single-agent** – The rover performs tasks independently.

iii. What are the actions the agent can take?

The Mars Rover can:

- Move forward, backward, left, and right
- Avoid obstacles
- Capture images and videos
- Collect rock and soil samples
- Analyze samples using onboard instruments
- Drill into rocks

Send data to Earth

iv. How can one evaluate the performance of the agent?

The rover's performance can be evaluated based on:

- Successful completion of exploration tasks
- Accuracy of scientific observations
- Number and quality of samples collected
- Safe navigation with minimal collisions
- Efficient power usage

v. What sort of agent architecture is most suitable for this agent? Why?

Most Suitable Agent: Model-Based Goal-Based Agent

Reason:

- Maintains a model of the Martian environment.
- Plans routes to reach scientific targets.
- Makes decisions using current sensor data and stored knowledge.

Avoids obstacles while achieving exploration goals.

3.

ANSWER: 8-Queens Problem – Problem Formulation and Search Space Problem

1. Initial State

- **Empty chessboard with no queens placed.**

2. State Space

- **All possible arrangements of 8 queens on the chessboard.**
- **A common representation is one queen per column, reducing the search space.**
- **Each state represents a partial or complete placement of queens.**

3. Actions (Successor Function)

At each step:

- **Place one queen in the next column.**
- **Choose a row where the queen is not attacked by previously placed queens.**

4. Goal Test

A solution is reached when:

- **All 8 queens are placed, and**
- **No two queens share the same row, same column, or same diagonal.**

5. Path Cost

- **Each queen placement has a cost of 1.**
- **Total cost for a complete solution = 8.**
- **Breadth-First Search (BFS): Explores level by level but requires large memory.**

- **Depth-First Search (DFS): Explores one branch completely before backtracking; commonly used.**
- **Backtracking Search: Places queens one by one and backtracks whenever a conflict occurs. It is the most efficient and widely used method for the 8-Queens problem.**

Example of a Valid Solution
Row: 1 2 3 4 5 6 7 8
Column: 1 5 8 6 3 7 2 4
This means the queens are placed at:

- **(1,1)**
- **(2,5)**
- **(3,8)**
- **(4,6)**
- **(5,3)**
- **(6,7)**
- **(7,2)**
- **(8,4)**

No two queens attack each other.

4.
ANSWER: OLA Cab Booking – Problem Solving Agent and Pseudocode

Type of Problem-Solving Agent

The **Goal-Based Agent** is the most suitable for this problem.

Reason:

- The customer's **goal** is to travel from the source location to the destination.
- The agent compares available cab options (Mini, Micro, Sedan, Shared, Prime, etc.) based on cost, comfort, availability, and travel time.
- It selects the cab that best satisfies the user's preferences and reaches the destination.

Problem Formulation

- **Initial State:** Customer is at the source location.
- **Goal State:** Customer reaches the destination.
- **Actions:**
  - Enter source and destination.
  - View available cab types.
  - Select a preferred cab.
  - Confirm booking.
  - Travel to the destination.
- **Path Cost:** Fare, travel time, and waiting time.
- **Pseudocode**

```
Algorithm: OLA Cab Booking

START
Input source location
Input destination
Display available cab types
(Mini, Micro, Sedan, Shared, Prime)
Customer selects preferred cab
IF selected cab is available THEN
    Display fare and estimated arrival time
    Confirm booking
    Assign nearest driver
    Driver picks up customer
```

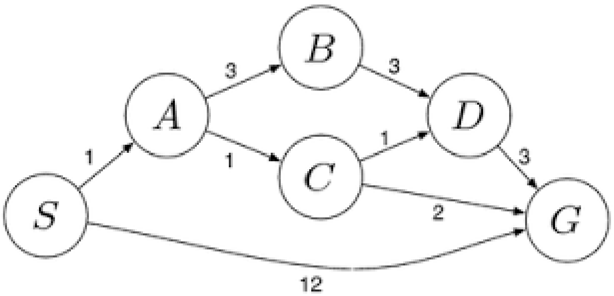
```
        Travel to destination
        Display "Trip Completed"
    ELSE
        Display "Selected cab not available"
        Suggest another cab type
    ENDIF

    STOP
```

5.

ANSWER: ***To solve the logistics problem using Uniform Cost Search (UCS), we find the path from S (Start) to G (Goal) with the minimum total cost. From the given graph, the edge costs are:***

Given Graph



Edge	***Cost***
S → A	***1***
S → G	***12***
A → B	***3***
A → C	***1***
B → D	***3***
C → D	***1***
C → G	***2***
D → G	***3***

Uniform Cost Search (UCS) Execution

Step 1

Start at S

Frontier	***Cost***
S	***0***

Step 2

Expand S.

Add neighbors: A(1), G(12).

Frontier	***Path Cost***
A	***1***
G	***12***

Step 3

Expand A(1).

Add B(4) and C(2).

Frontier	***Path Cost***
C	***2***
B	***4***

Frontier Path Cost

G	***12***

Step 4
Expand C(2).
Add D(3) and G(4).

Frontier Path Cost

D	***3***
B	***4***
G	***4***
G	***12***

Step 5
Expand D(3).
Add G(6).

Frontier Path Cost

B	***4***
G	***4***
G	***6***
G	***12***

Step 6
The next node with the lowest path cost is G(4). Since the goal is selected for expansion, UCS stops.
Optimal Path
S → A → C → G
Total Cost
4